

INDOOR NAVIGATION SYSTEM WITH OBJECT & FACE IDENTIFICATION FOR VISUALLY IMPAIRED INDIVIDUALS

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Abstract— Indoor navigation is an important issue in large buildings where GPS is weak or not available, especially for individuals who are blind or visually impaired. Smarter help for safe indoor movement is now possible thanks to advancements in computer vision. Similar to a person without a handicap, visually impaired users frequently lack the capacity to recognize people in their visual surroundings, notice impediments, and navigate new indoor places, all of which might lower their level of independence and social confidence. While current aids, like a guiding dog or white cane, offer assistance, most of them are inadequate in intricate indoor settings. The goal of this project is to offer voice-based navigation in real time, local items that are visible to the human eye, and facial recognition for recognized people. In order to facilitate long-term contact with assistance, unknown faces will also be kept in the database for previously unidentified personal identification. The suggested project makes use of an application that interfaces with CNN-based facial recognition, YOLO-based object detection, and A*-based floor-map navigation. Python APIs process the camera input in real-time, and the mobile application is based on React Native-Expo. It aims to be a broadly accessible, affordable, and portable assistive device.

Keywords: *Augmented Reality (AR), Indoor Navigation System, Voice Assistant (Aria), Speech-to-Text (AssemblyAI), Natural Language Processing, Computer Vision, Object Detection (YOLOv5/MobileNet), Face Recognition (FaceNet/ML Kit), Assistive Technology, Visually Impaired Navigation, Real-Time Systems, Cloud-Based Processing, Human-Computer Interaction*

I. INTRODUCTION

Navigation within indoor environments remains a significant challenge, particularly for visually impaired individuals. Unlike outdoor navigation systems that rely on Global Positioning System (GPS) technology, indoor spaces lack reliable positioning systems, making it difficult for users to move independently and safely. This limitation often results in reduced mobility, dependence on others, and increased risk of accidents in unfamiliar environments such as hospitals, shopping malls, and educational institutions.

With recent advancements in Augmented Reality (AR), Artificial Intelligence (AI), and voice-based interaction technologies, there is a growing opportunity to develop intelligent assistive systems that can address these challenges. AR enables the overlay of digital information onto real-world environments, providing intuitive visual guidance, while AI techniques such as computer vision allow real-time understanding of surroundings.

Additionally, voice assistants have transformed human-computer interaction by enabling natural and hands-free communication.

This project proposes an AR-based indoor navigation system integrated with a voice assistant (Aria) to enhance accessibility for visually impaired users. The system combines multiple technologies, including cloud-based Speech-to-Text (STT), object detection, and face recognition, to provide a comprehensive assistive solution. Users can interact with the system using voice commands, which are processed using online STT services to determine their intent. Based on the request, the system provides navigation guidance, detects obstacles in real time, and identifies known individuals.

The navigation component utilizes AR to display directional cues, while deep learning models such as YOLOv5 or MobileNet are employed for object detection. Face recognition technologies such as FaceNet or ML Kit further enhance situational awareness. The system delivers feedback through pre-recorded voice assist audio or Text-to-Speech, ensuring effective communication.

The primary objective of this system is to improve independence, safety, and confidence among visually impaired individuals by providing an integrated, real-time navigation and assistance platform. This approach demonstrates the potential of combining AR and AI technologies to create inclusive and intelligent assistive solutions.

Main Contributions:

1. To provide safe indoor navigation through real-time voice guidance: The system uses computer vision to map the surroundings and create a virtual path. It guides users with continuous voice instructions and alerts them of any obstacles in their path, ensuring safety during indoor movement.
2. To enable identification of known individuals using facial recognition: The system detects faces in real time and informs users when familiar people are nearby, promoting confident and natural social interaction.

3. To store and manage unknown face data for future recognition: When the system detects unfamiliar faces, it saves their data for future reference, helping users build familiarity and improving long-term interaction.

II. LITERATURE SURVEY

Juile J et al. (2024) proposed a low-cost smartphone-based augmented reality indoor navigation system using ARCore, Unity, and NavMesh. The system applies SLAM to build a 3D environment model, track user location, and provide real-time route updates through AR overlays. NLP enables voice interaction for accessibility. Although effective, it faces screen jitter and requires initial calibration. Future work includes GPS integration and improved tracking.

Wensheng Zhang et al. (2020) proposed a low-cost smartphone-based indoor navigation system using BIM and AR for GPS-denied environments. A 3D BIM model is converted into a 2D map, while AR nodes enable camera-based positioning. Their IARA algorithm selects routes considering distance, visibility, and AR node usage. Tests showed improved usability and engagement, though setup requires manual AR node placement and BIM model development time.

R.G. Baldovino et al. (2024) developed a smart indoor assistive system for visually impaired users using CCTV input, YOLO-based object detection, and audio-haptic feedback. It alerts users through sound or vibration when objects are within 0.3 meters. The model achieved 0.92 mAP and was deployed as “Project MATA” using Django. Although effective and low-cost, it faced slight latency, reduced edge precision, and hardware-dependent performance.

Izzat N. Jarudi et al. (2023) studied how face recognition accuracy decreases with distance. Using Gaussian blur, they tested human participants on full, internal, and external facial features. Full-face images gave the best recognition, while isolated features reduced performance. They also trained AlexNet CNN models on the FaceScrub dataset, which showed similar results. The study concluded that holistic spatial integration of facial features is essential for recognizing faces at a distance.

R. M. S and A. M. R et al. (2023) proposed a smartphone-based AR indoor navigation system using ARCore, Flutter, and Firebase for GPS-denied environments. It provides markerless navigation with virtual arrows and real-time route updates. Firebase stores navigation paths, while ARCore handles tracking and mapping. The system showed effective navigation but faced battery drain, device compatibility issues, and internet dependency. Future work includes hybrid navigation and improved security.

R. Goud et al. (2022) proposed a low-cost AR-based indoor navigation system for GPS-denied spaces using SLAM and image recognition. Users scan surroundings for real-time localization, and the A* algorithm computes the shortest route displayed as AR overlays. Built with Unity 3D, ARCore/Vuforia, and Firebase, it achieved smooth navigation without special hardware. However, performance depends on lighting, repetitive structures, and device capability. Future work includes obstacle detection and AI integration.

S. C. J. et al. (2020) proposed an AR-based indoor navigation system using ARCore SLAM for localization and QR codes for position reset. Unity NavMesh and the A* algorithm compute obstacle-aware optimal routes, displayed through AR arrows and a real-time minimap. The system achieved high tracking accuracy (~0.08 m error) without external hardware. However, it depends on lighting, AR stability, and battery life. Future work includes

dynamic obstacle detection and support for visually impaired users.

Rakesh Chandra Joshi et al. (2020) proposed an assistive system combining YOLO-v3 with a DSP processor, camera, and ultrasonic sensors for real-time obstacle detection and audio feedback. A labeled and augmented dataset was created using LabelImg. The trained YOLO-v3 model achieved 95.19% detection accuracy and 99.69% recognition accuracy. Compact audio outputs reduced processing delay, and the system was tested using DSP hardware with an 8 MP camera.

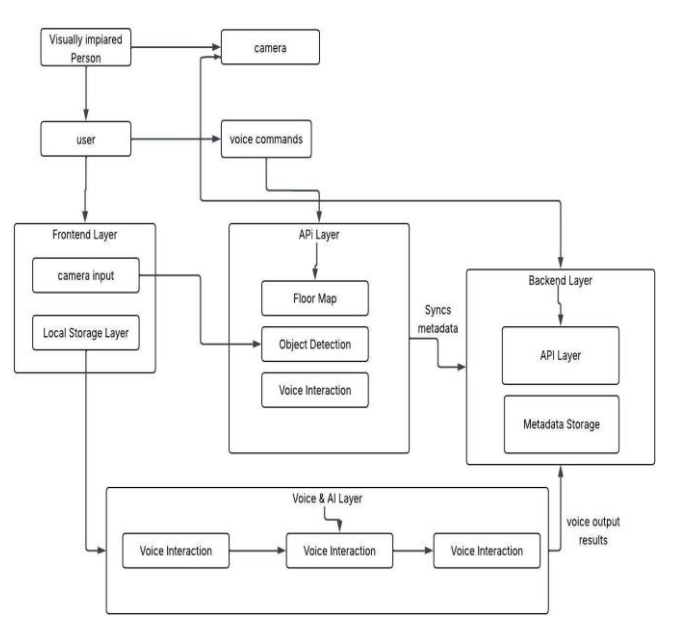
III.SYSTEM DESIGN

The architecture schematic shows the general layout of the Indoor Navigation System with Object and Face Identification for Visually Impaired Individuals and the interactions between its key components.

The suggested system design is shown in detail in Fig. 1. To assist visually impaired persons in securely navigating indoor settings, the system makes use of voice interaction, facial recognition, and real-time object identification. The user interacts with the system through voice commands and the mobile device's camera, which captures live footage of the surroundings. The Frontend Layer, developed with React Native Expo, controls camera input and stores local data, such as identifiable faces and floor maps, for offline use.

The API Layer, which consists of three lower-level modules, then uses the data: the Voice/I/O module, which is in charge of identifying commands and system feedback, the Object Detection module, which employs computer vision models like YOLO, and the Navigation module, which displays the floor maps. By storing metadata (information about the user), synchronizing maps and face data, and sending service requests from the mobile app to the server, the Backend Layer supports the proper operation of the API layer. Hands-free navigation is made possible by the Voice and AI Layer, which combines text-to-speech processing and speech recognition. Ultimately, these three layers guarantee smooth data flow, intelligent environment awareness, and real-time auditory guidance and feedback, making indoor mobility safer and more accessible for people with visual impairments.

Fig1: System Architectural diagram



Dependencies are highlighted in the diagram when a class depends on another to carry out its operation. The suggested class diagram is shown in detail in Fig. 2. The user initiates commands that the MobileApplication receives, and the BackendServer also manages the user's preferences. The VoiceAssistant communicates output to the user, the ObjectDetection module senses the surroundings, and the NavigationModel computes paths for the MobileApplication. Situational awareness depends on the fundamental sensory and cognitive modules, ObjectDetection and FaceRecognition. To load and store their individual machine learning models and map data, they rely on LocalStorage (using +cacheModels and +cacheMaps). The VoiceAssistant receives instructions from the NavigationModel and utilizes functions like +convertTextToSpeech() to produce the user's final audio assistance. In conclusion, the diagram shows a tightly integrated voice-controlled system in which the MobileApplication coordinates location and input data, transmits it to the Detection/Recognition modules for environmental analysis and the NavigationModel for path computation, and then uses the VoiceAssistant to deliver instructions.

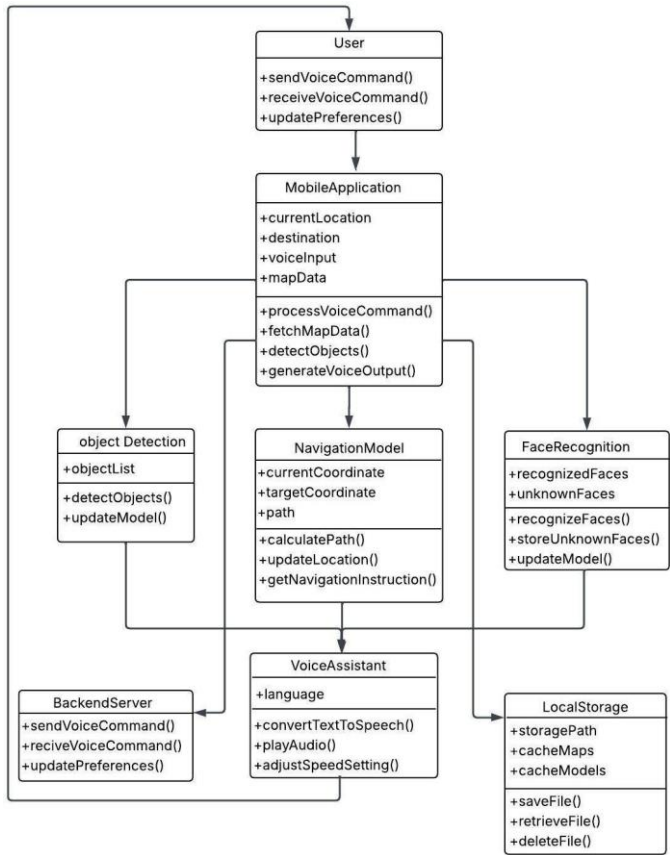


Fig. 2:Class Diagram

displaying the Administrator and the Visually Impaired User as the two main protagonists. The suggested use case diagram is shown in detail in Fig. 3. The operational flow is driven by the visually impaired user, who first accesses the system's essential features via the Start Application use case. The primary objective, Indoor Navigation, is intrinsically reliant on three additional features: it "includes" Voice Feedback to deliver auditory instructions, and it is closely linked to Object Detection for real-time safety alerts about obstacles and Facial Recognition for identifying people in the vicinity, both of which depend on Voice Feedback for communication. In order to maintain the accuracy and timeliness of the underlying map data utilized for navigation, the Administrator is solely in charge of the Manage Floor Maps use case. This structure effectively separates the day-to-day user experience features from the essential system maintenance functions.

The system's coordination of various modules to serve the user is made evident by the Sequence Diagram. The suggested sequence diagram is shown in detail in Fig. 4. The user chooses a mode on the mobile app interface to start the process. This results in a center Conditional Flow that controls the order that follows. The Mobile App Interface submits a Route Request to the Navigation System during the Navigation Flow (If Mode = Navigation). After retrieving map data from the Floor Map Database, the Navigation System calculates the optimal path using the pathfinding algorithm and transmits the resulting instructions to the Voice Assistance System. The user is then given spoken guidance by the voice assistance system. The Detection Flow operates continually in tandem with navigation, whereby the camera continuously feeds the Object Detection and Facial Recognition modules with Live Video Feed. After

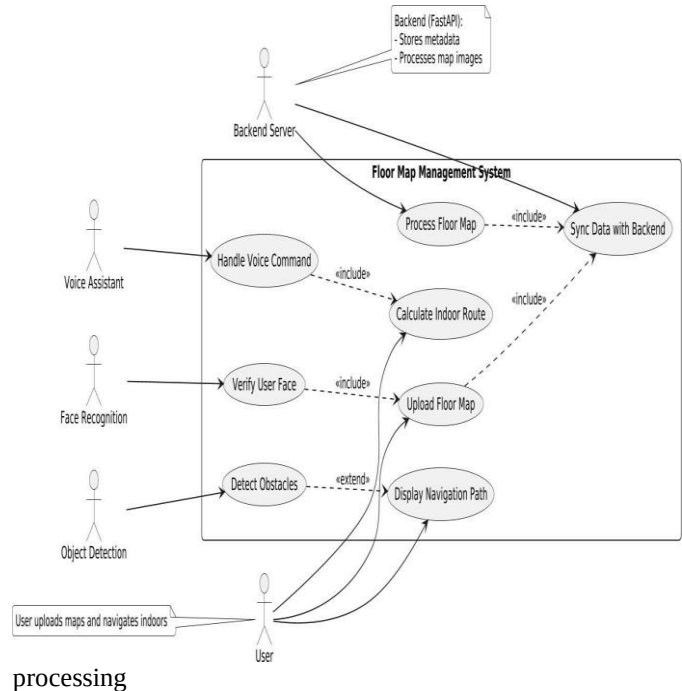


Fig3:Use Case Diagram

The Object Detection module continually analyzes each frame of the live video stream from the mobile camera in order to identify potential threats, close objects, and barriers in the user's environment. This involves identifying typical interior features including tables, seats, walls, doorways, stairwells, and moving objects like people. The module classifies an object or obstruction and calculates its estimated position and distance from the user after it has been detected. The Object Detection system creates Alert Data based on this analysis, which includes details about the kind of object found and the potential danger it poses.

The Voice Assistance System receives this alert data instantly and translates it into a clear auditory warning, such as "Obstacle ahead," "Person on the left," or "Stairs in front." This guarantees that the user gets voice feedback in real time and may move quickly to prevent crashes or hazardous situations. Similar to this, the Facial Recognition module looks for human faces in the surroundings by analyzing the same live video feed. Key facial traits are extracted by the module when a face is recognized in the camera frame, and they are then sent to the Face Recognition Database for comparison with previously saved face profiles. In the event that a match is discovered, the system notifies the Voice Assistance System and verifies the individual's identity. In order to help the user identify familiar people and enhance social contact, the Voice Assistance System then broadcasts the name of the recognized person to the user. However, the module creates an Unknown Face Alert

and notifies the Voice Assistance System if the identified face does not match any record in the database.

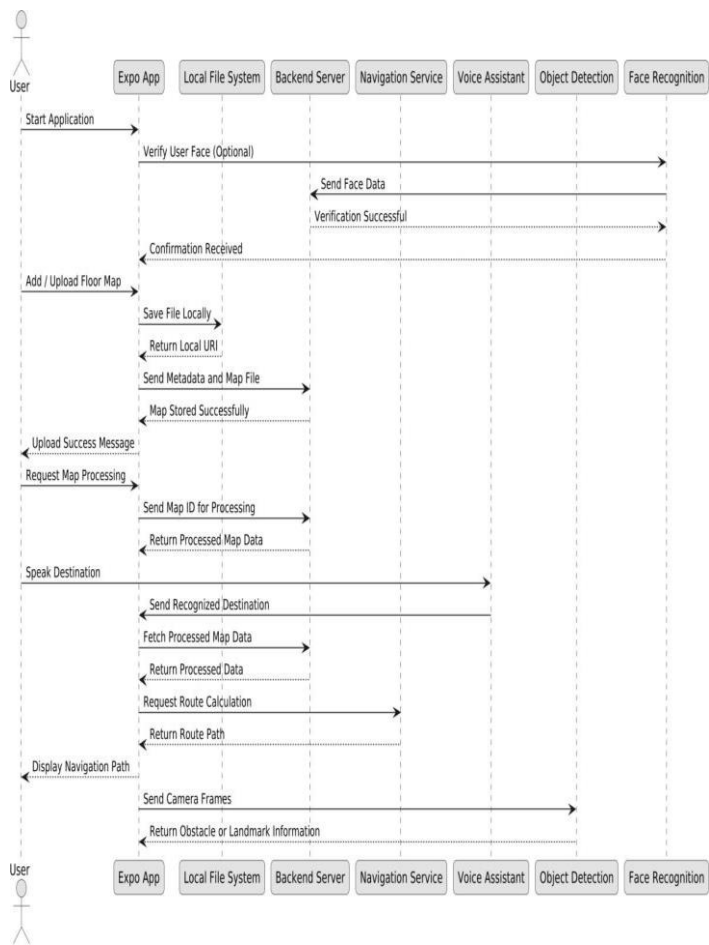


Fig .4:Sequence diagram

IV. METHODOLOGY

The Indoor Navigation System's Level 1 Data Flow Diagram shows how the main functionality is broken down into three interconnected processes that communicate with two data storage and the user. The suggested data flow diagram is shown in detail in Fig. 5. A voice command from the user to the mobile app interface (Process 1) starts the cycle. As the command interpreter, this interface converts the command into an activation instruction delivered to the Recognition Module (Process 3) or a Navigation Request submitted to the Navigation System (Process 2).

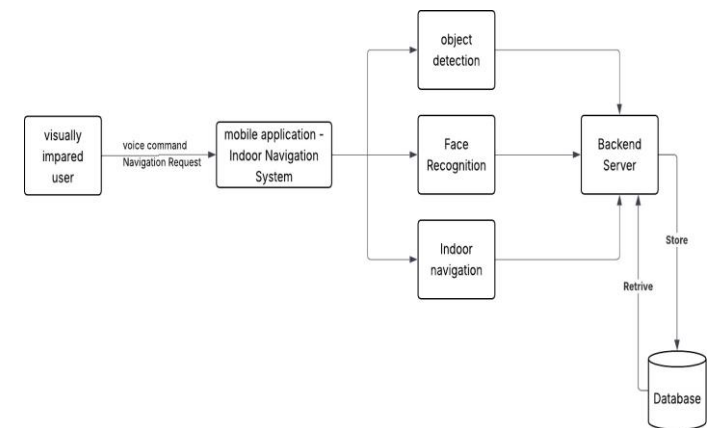


Fig.6:Data Flow Diagram

In order to compute a path, the Navigation System obtains Map Data from the Floor Map Database (Data Store 1) and generates Navigation Instructions, which are then sent to the user as Voice Guidance. Concurrently, the Recognition Module functions independently, obtaining Live Camera Feed to conduct visual analysis against the Face Recognition Database (Data Store 2), resulting in two real-time outputs that are returned to the user: Real Time Obstacle Alerts for safety and Face Recognition. outcomes for the social setting. The system's dual, integrated role of goal-oriented navigation and ongoing environmental awareness is highlighted throughout this entire structure.

V. EXPERIMENTAL SETUP

The experimental arrangement of the suggested system was carried out in an organized way to assess the functionality and effectiveness of every combined module. The execution took place within a Windows-based development environment, employing Python for backend processing and using React Native for developing the mobile application.

A. Voice Assistance Module:

The speech processing module was first developed using the AssemblyAI cloud-based Speech-to-Text (STT) API, allowing precise transcription of user voice commands. The system underwent testing with both pre-recorded audio and live microphone recordings to assess transcription accuracy, response time, and reliability across different speech conditions.

B. Navigation Module:

The navigation module was tested by assigning source and destination points within a mapped indoor environment. As illustrated in Fig-8.4, the user selects the starting point and destination. The navigation process begins as shown in Fig-8.5, where the system generates a navigation path using the Immersal SDK. During movement, the system continuously updates the navigation guidance, as depicted in Fig. No. 8.6, ensuring accurate and real-time navigation through AR overlays.



Fig 7: Assigning the source and the destination.

C. Object Detection Module:

The module for object detection was tested by obtaining real-time camera footage and identifying obstacles in the surroundings. As illustrated in Fig-8, the system effectively recognizes items like furniture and delivers suitable notifications. The analysis of the performance focused on detection accuracy and response time.

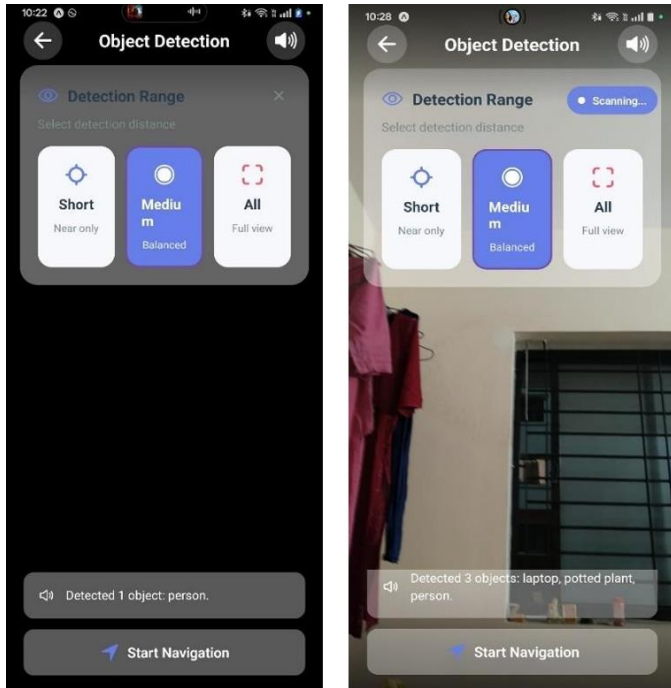


Fig 8: Detecting the object.

D. Face Recognition Module:

The face recognition system was evaluated by saving and identifying familiar people. As shown in Fig. No. 8.8, facial information was recorded and saved for recognition. The system underwent assessment across various conditions, such as changes in lighting and user positioning, to guarantee dependable recognition performance.

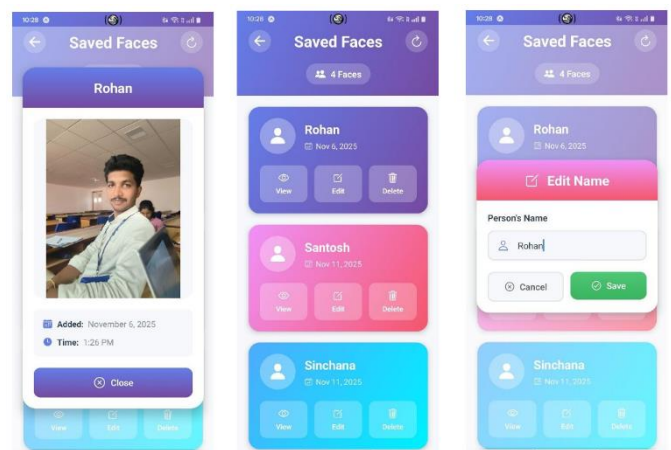


Fig 9: Saving the face for future recognition.

The response system was established using pre-recorded voice assistant audio (Aria) combined with optional Text-to-Speech (TTS) capabilities. The system underwent tests in both calm and loud settings to guarantee efficient and distinct communication. Furthermore, haptic feedback was taken into account to improve accessibility in difficult acoustic environments.

Ultimately, the comprehensive system performance was assessed based on these parameters:

- Precision of speech recognition.
- Time taken for system response.
- Efficacy of AR-driven navigation.
- Dependability of object identification.
- Precision of facial recognition.

The experimental assessment shows that the suggested system offers dependable real-time support and greatly improves navigation effectiveness and situational awareness for users with visual impairments.

VI. RESULT AND DISCUSSION

The suggested AR-driven indoor navigation system combined with a voice assistant showed efficient performance in all functional areas. The speech recognition feature offered precise and instantaneous transcription of user commands, allowing seamless interaction. The navigation module effectively created and refreshed indoor routes, providing dependable direction and enhancing user movement. The obstacle detection system effectively recognized hazards instantly, improving user safety by delivering prompt notifications. In the same way, the face recognition system delivered reliable results in recognizing familiar people across different situations. The feedback system, designed with voice assist audio and optional text-to-speech, provided users with clear and comprehensible responses. Despite some minor limitations like decreased accuracy in noisy settings and minor delays from cloud processing, the overall system was found to be dependable, efficient, and successful in enhancing navigation and situational awareness for visually impaired individuals.

VII. CONCLUSION

The Navigation System with Object Detection and Face Recognition project combines cutting-edge computer vision technology to deliver an intelligent, interactive, and intuitive indoor navigation experience. The system improves safety, accessibility, and customisation by fusing face recognition with real-time object detection. It helps users navigate inside spaces while instantly recognizing known people and barriers. Accurate navigation support is ensured by the combination of environment mapping, YOLO-based object identification, and facial recognition modules, which makes it helpful in complicated locations like shopping centers, workplaces, and educational institutions. All things considered, the research shows how artificial intelligence may produce navigation solutions that go beyond straightforward direction advice. A safer and more inclusive navigation ecosystem is made possible by the foundation it offers for upcoming improvements like voice assistance, gesture control, cloud storage, and IoT connectivity.

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